$F = 400 \text{ lbf}$

$\delta = .009 \text{ in}$

$\varepsilon = 10007604 \text{ Psi}$ (6061 T6 Aluminium)

$\downarrow$

$10 \times 10^6 \text{ Psi}$

$\delta = \frac{Fl}{\varepsilon A}$ $\quad \underset{\text{in}}{(.009)} = \frac{(400)(L)}{(\varepsilon)(A)}$

$$L = \frac{(\text{delf})(\varepsilon)(A)}{400}$$

$5.946 \times 10^7 \text{ Pa}$

$\hookrightarrow$ 8623.94 Psi or 8.6 Ksi < 40 Ksi

Lowest F.S is 4.62514

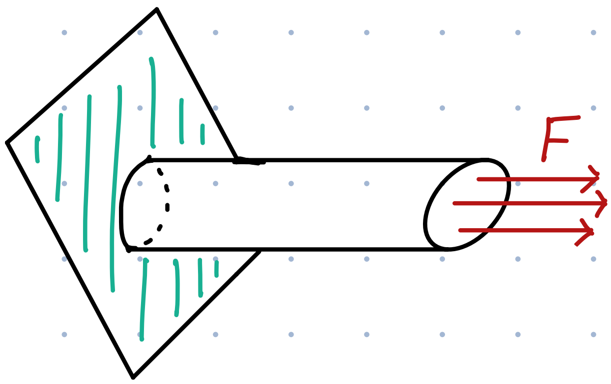
$.009 = \frac{(400)(L)}{\varepsilon A}$ $\quad A = \left(\frac{d}{2}\right)^2 \pi$

$d = .25 \text{ in}$

$\hookrightarrow$ $\quad A = .0491 \text{ in}^2$

$$\frac{400(L)}{(10 \times 10^6)(.0491)} = .009$$

$L = 11.0475 \text{ in}$



Known

$F = 400\ lbf$

$\delta_{max} = .009$ in

$E = 10007604$ psi (6061 T6 Aluminum)

$\Rightarrow 10\times10^6$ psi

Unknown

L, A, ϕ

$\phi = .25$ in $\rightarrow$ $A = \left(\frac{.25}{2}\right)^2 \cdot \pi \rightarrow = .0491\ in^2$

$\delta = \frac{FL}{EA}$ ↳ Solve for L $\quad L = \frac{\delta E A}{F}$ Solving numerically

$$L = \frac{\overset{in}{(.009)}\,\overset{\cancel{lb}/\cancel{in^2}}{(10\times10^6)}\,\overset{\cancel{in}^2}{(.0491)}}{400\ (\cancel{lb})}$$

$\Rightarrow 11.0475$ in

Max stress $= 5.946\times10^7$ Pa ↳ 8623.94389 psi or 8.6 ksi

Converted to psi

$S_y = 40$ ksi

$8.6 < 40 \rightarrow$ ✓ The max stress is lower the yield stress.

Horray! Yippe!

Solidworks – 0.2285 mm $\rightarrow$ .0089961 in

Hand Calc. – .009 in

% difference = 0.04334%

$$\delta = \frac{(400)(11.0475)}{(10\times10^6)(.0491)} = 9\times10^{-3}\ in$$

$$P = \frac{|.009 - .0089961|}{(.009 + .0089961)(\frac{1}{2})}(100) = .04334\% \rightarrow \text{Basically none.}$$

$$\frac{.25}{11} = = .02273 \qquad K_t \approx 2.8 \qquad \sigma_{max} = K_t \cdot \sigma_{nom}$$

$$\sigma = (2.8)(8133.716)$$

$$\sigma_{max} = 22774.4048 \text{ Psi}$$

22.774 ksi

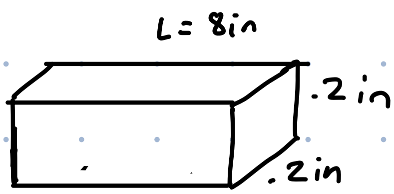
S.F = 1.756

load → 450

Thick → .2 in

$$L = \frac{(.009)(10\times10^6)(.2^2)}{450}$$

$\Rightarrow$ 8 in



(not to scale)